Marmara Lectures

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Discrete signals

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Before going on with deicrete signals, let us remember what continuous signals are: Continuous signals are defined on real numbers $\mathbb{R}$. Notationally they are denoted by small case letters and indexed by curly braces and with a dependent variable t, like $x(t)$, $y(t)$. As $t \in \mathbb{R}$, $x(2.7)$, $y(\sqrt{3})$, $x(\pi)$ are well-defined.

Discrete signals are defined on integers $\mathbb{Z}$. Notationally they are they are denoted by capital letters and indexed by angle braces and with a dependent variable n, like $X[n]$, $Y[n]$ etc. As $n \in \mathbb{Z}$, $X[2]$, $Y[-4]$, $X[0]$ are well-defined. But $X[2.7]$, $U[\sqrt{3}]$, $W[\pi]$ are undefined and using such expressions indicates a mistake.

Discrete exponential signals

Exponential discrete signals are defined as follows.

$$X[n] = Ae^{\alpha n}, n \in \mathbb{I}$$

The python code which will print this discrete signal for $\alpha = -0.1$ is given below.

```
1.  import matplotlib.pyplot as plt
2.  import numpy as np
3.  n = np.arange(0,0.20,1);
4.  X = np.exp(-0.1*n);
5.
6.  plt.stem(X)
7.  plt.xlabel('n')
8.  plt.ylabel('X[n]')
9.
10. plt.show()
```

And, this is the resulting plot:



Discrete sinusoid signals

Exponential sinusoids are defined as

$$X[n] = A\sin(\Omega n + \Omega_0),$$

where $n \in \mathbb{I}$ and $\Omega, \omega_0 \in \mathbb{R}$.

A plot for $\Omega = \frac{2\pi}{10}$, $\Omega_0 = 0$ is given below:



Discrete dirac deltas (Kronecker Deltas)

A kronecker delta is the analogue of a dirac delta in discrete domain.

$$\delta[n] = \begin{cases} 1, & \mathrm{n} = 0 \\ 0, & \mathrm{otherwise} \end{cases}$$

```
import matplotlib.pyplot as plt
import numpy as np
n = np.arange(-10,10,1);
X = (n==0);
plt.stem(X)
plt.xlabel('n')
plt.ylabel('X[n]')
plt.show()
```



Discrete unit step

A discrete unit step is described as

$$U[n] = \begin{cases} 0, & n < 0 \\ 1, & 0 \leq n \end{cases}$$

Discrete indicator function

A discrete indicator step can be described as

$$\chi_{[a,b]}[n] = \begin{cases} 1, & a \leq n \leq b \\ 0, & \text{otherwise} \end{cases}$$

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